

BUSINESS ANALYTICS

INDIVIDUAL CASE STUDY

Predicting Road Accident Severity Using Location, Time, and Weather Factors

Name: Kanish Kumaran M

Roll No: CB.SC.U4CSE23424

Class: CSE E

Domain: Transportation / Public Safety

Executive Summary

Road-accident severity is influenced by contextual factors such as location, time, weather, road environment, vehicle involvement and collision characteristics. This case study develops a Business Analytics workflow to identify such patterns and demonstrate a preliminary classification approach using publicly accessible accident-reporting webpages.

Data were collected through Google News scraping on the Apify platform. The initial collection consisted of 1,000 raw article rows across multiple batches, with 787 unique URL values in that initial combined file. Subsequent collection runs expanded the current working article corpus to 848 records. Because multiple news reports can describe the same accident, these records are treated as an article-level source corpus rather than as 848 confirmed unique accidents.

The analysis extracts state/location evidence, time-of-day evidence, reported weather indicators, road context, vehicle involvement and collision indicators. A validated pilot modeling file contains 290 records with non-unknown severity and clear single-state evidence. Four classifiers were evaluated: Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

In the saved pilot experiment, the Decision Tree achieved the highest macro F1 among the tested models at 0.391, with accuracy of 0.414. These values are preliminary article-level pilot metrics and are not claimed as final accident-event predictive performance.

The major analytical finding is therefore twofold: the scraped reports contain useful contextual evidence for severity analysis, but event-level deduplication and validation are essential before the model can be used as a reliable operational road-safety predictor.

Report at a Glance

Item	Final report value
Collection method	Public Google News pages collected through Apify
Initial combined raw rows	1,000
Unique URLs in initial combined file	787
Current working article corpus	848 records
Pilot modeling dataset	290 records
Severity classes	Fatal / Serious / Minor/Reported Injury
Models evaluated	Logistic Regression, Decision Tree, Random Forest, Gradient Boosting
Highest pilot macro F1	0.391 (Decision Tree)
Highest pilot accuracy	0.414 (Decision Tree)

1. Problem Statement and Objectives

Road accidents do not have the same consequences across all locations and circumstances. Differences in time of occurrence, weather, road context, vehicle involvement and collision characteristics may be associated with differences in accident severity. For transportation and public-safety decision making, identifying these patterns can support the prioritization of road-safety reviews, warnings, patrol planning and infrastructure interventions.

The case study therefore treats accident severity as a classification problem using structured indicators extracted from publicly available accident reports.

Objectives

1. Collect publicly accessible road-accident reports through web scraping and document the collection procedure.
2. Prepare and explore the collected information to identify location, time, weather, vehicle and collision patterns associated with reported severity.
3. Implement and compare classification models and translate the findings into practical road-safety and resource-planning insights.

Scope and Analytical Unit

The current dataset is an article-level source corpus. One accident may be reported by several news organizations, so the number of articles is not interpreted as the number of unique accidents. The pilot model uses only records that passed conservative severity and location-evidence filters. Death and injury counts are not used as predictors because they directly reveal or define severity and would create target leakage.

2. Data Collection and Dataset Description

The primary data source was publicly accessible Google News results and the linked news articles. Apify was used to automate collection, resolve article URLs and retrieve article bodies where available. The collection used accident-focused search queries such as road-accident, road-crash and fatal-road-accident combinations with location terms.

Collection stage	Records / evidence	Purpose
Initial combined batches	1,000 raw rows	Base scraped corpus
Initial URL deduplication	787 unique URL values	Remove repeated article URLs
Later collection runs	Additional successful 50/100/200/500-result runs	Expand coverage
Current working corpus	848 article records	Working source for EDA
Pilot modeling dataset	290 records	Validated subset used for the saved model experiment

The Apify run history and run log below provide evidence of the collection process. The screenshots are included to document the actual scraping activity, consistent with the requirement that web-scraped data collection procedures be described.

Status	Actor	Task	Results	Usage	Started	Finished	Duration	Build	Origin
☒ Stored 500 articles.	Google News Scraper A...	auto...er-api @ Pay per event	500	\$2.63	2026-09-24 23:35:57	2026-09-25 00:00:07	24 m 10 s	0.1.12	Web
☒ Stored 200 articles.	Google News Scraper A...	auto...er-api @ Pay per event	200	\$1.05	2026-09-24 23:01:38	2026-09-24 23:10:44	9 m 5 s	0.1.12	Web
☒ Actor succeeded with 50 results in the dataset	Google News Scraper · F...	memo...craper @ Pay per event	50	\$0.22	2026-09-24 22:50:46	2026-09-24 22:54:01	3 m 14 s	0.1.41	Web
☒ Actor succeeded with 50 results in the dataset	Google News Scraper · F...	memo...craper @ Pay per event	50	\$0.22	2026-09-24 22:38:52	2026-09-24 22:44:05	5 m 12 s	0.1.41	Web

Figure 1. Apify run history showing successful Google News scraping batches

```

2026-09-24T16:48:02.443Z [INFO] excludeWords: [],
2026-09-24T16:48:02.445Z [INFO] sitefilter: []
2026-09-24T16:48:02.446Z [INFO]
2026-09-24T16:48:02.448Z [INPUT] 111 resolved + query=111 topic=0 publisher=0 cluster=0
2026-09-24T16:48:02.639Z [INFO] [CACHE] loaded 91 decoded URLs from KV store "google-news-url-cache"
2026-09-24T16:48:02.640Z [INFO] [INTERNAL] starting Google News pipeline | inputs=111 maxItems=50 resolveUrls=true enrichBody=true extractEntities=true
2026-09-24T16:48:03.737Z [INFO] [RSS] query (road accident in Chennai Tamil Nadu) | 20 eligible items (of 100)
2026-09-24T16:48:06.190Z [INFO] [DECODE] query (road accident in Chennai Tamil Nadu) | batch decoded 5/5 fresh (cache hits this feed: 15)
2026-09-24T16:49:20.967Z [INFO] [RSS] query (road crash in Chennai Tamil Nadu) | 20 eligible items (of 78)
2026-09-24T16:49:26.369Z [INFO] [DECODE] query (road crash in Chennai Tamil Nadu) | batch decoded 15/15 fresh (cache hits this feed: 5)
2026-09-24T16:50:30.775Z [INFO] [RSS] query (fatal road accident in Chennai Tamil Nadu) | 20 eligible items (of 78)
2026-09-24T16:50:35.247Z [INFO] [DECODE] query (fatal road accident in Chennai Tamil Nadu) | batch decoded 15/15 fresh (cache hits this feed: 5)
2026-09-24T16:51:00.930Z [INFO] [CACHE] saved 126 decoded URLs to KV store
2026-09-24T16:51:01.005Z [INFO] [INTERNAL] DONE | rows=50 (rss-ok=3 rss-failed=0 url-resolved=50/50 cache-hits=25 body-ok=39/50) inputs: query=3 topic=0 publisher=
2026-09-24T16:51:01.009Z [MAIN] stats: {
2026-09-24T16:51:01.010Z   rowsEmitted: 50,
2026-09-24T16:51:01.013Z   rowsFailed: 0,
2026-09-24T16:51:01.014Z   rssFetched: 3,
2026-09-24T16:51:01.016Z   rssFailed: 0,
2026-09-24T16:51:01.018Z   urlResolutionAttempted: 50,
2026-09-24T16:51:01.020Z   urlResolutionOk: 50,
2026-09-24T16:51:01.021Z   urlCacheHits: 25,
2026-09-24T16:51:01.023Z   bodyResolutionAttempted: 50,
2026-09-24T16:51:01.025Z   bodyResolutionOk: 39,
2026-09-24T16:51:01.026Z   cfbypassAttempted: 0,
2026-09-24T16:51:01.028Z   cfbypassOk: 0,
2026-09-24T16:51:01.030Z   queryInputs: 3,
2026-09-24T16:51:01.032Z   topicInputs: 0,
2026-09-24T16:51:01.034Z   publisherInputs: 0,
2026-09-24T16:51:01.035Z   clusterInputs: 0
2026-09-24T16:51:01.037Z }

```

Figure 2. Apify run log showing accident queries, URL resolution and article-body extraction

2.1 Collected Dataset Output

Analyze Road Accident Propo...

Run #KMSTWAIhXwgyCbG

Ask Gemini

console.apify.com/actors/runs/KMSTWAIhXwgyCbG#output

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KANISHKUMA...

Personal

Q Search...

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Apify Store

Dashboard

Actors

Runs

Tasks

Integrations

Schedules

Development

My Actors

Insights

Messaging

Proxy

Storage

RAM 0 MB / 16 GB

Usage \$0.88 / \$5.00

Upgrade to Starter

#	Author author	Body body	Body Extracted body:extracted	Body HTML body:html	Cluster Aggregate cluster:Aggregate
14	Express News Service	MADURAI: Five members of a family were killed after the car they were...	✓	null	null
15	null	null	✗	null	null
16	Rudhran Baraasu	CHENNAI: Tamil Nadu continued to record the highest number of road...	✓	null	null
17	The Hindu Bureau	The car that plunged into a gorge on Manjoor - Mulli road in Karamadai...	✓	<div id="readability-page-1" class="page"><div><div><picture><!--[[if IE 9]]><video style="display: none;"><!--endif--><source media="(min-width: 1600px)" sizes="960px" srcset="https://th-i.thgim.com/public/incoming/4bd4d4/article71325676.ece/alternates/LANDSCAPE_1200/Mulli%20accident%201.jpeg...><source...	null
18	Sivaraman R 10065	Chennai Police Commissioner A. Amalraj honouring a constable...	✓	<div id="readability-page-1" class="page"><div><div><picture><!--[[if IE 9]]><video style="display: none;"><!--endif--><source media="(min-width: 1600px)" sizes="960px" srcset="https://th-i.thgim.com/public/news/national/rf7tqt/article71263623.ece/alternates/LANDSCAPE_1200/amalraj.jpg"><source...	null
19	null	null	✗	null	null

apify

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Figure 3. Apify dataset output showing collected article fields and extracted article text

The output contains fields such as author, article body, publication information, title, source, URL and related metadata. The collected text was used as the basis for conservative extraction of structured analytical indicators.

☐	☒ Stored 100 articles.	 Google News Scraper API - ... autom...er-api @ Pay per event	100	\$0.56	2026-09-25 00:34:01	2026-09-25 00:39:27	5 m 25 s	0.112	Web
☐	☒ Stored 100 articles.	 Google News Scraper API - ... autom...er-api @ Pay per event	100	\$0.52	2026-09-25 00:20:31	2026-09-25 00:24:07	3 m 35 s	0.112	Web

Figure 4. Additional successful 100-article Apify runs

2.2 Important Variables

Variable	Meaning	Role
article_title	Headline of the collected report	Source/text
published_date	Publication date of the article	Temporal reference
state	State/location evidence from article text	Predictor
time_of_day	Morning, afternoon, evening or night evidence	Predictor
rain_or_wet	Explicit rain or wet-road evidence	Predictor
fog	Explicit fog/mist evidence	Predictor
highway	Explicit highway/expressway evidence	Predictor
bus / lorry / car / two-wheeler	Reported vehicle involvement	Predictor
head_on / rear_end / collision	Reported collision evidence	Predictor
severity	Fatal, Serious or Minor/Reported Injury	Target

No personal or confidential information was intentionally retained as an analytical variable. The report focuses on publicly reported accident characteristics rather than identifying individual people.

3. Data Preparation and Exploratory Analysis

3.1 Data Preparation

4. Combined the collected scraping batches into a common tabular structure.
5. Checked URL-level duplication and retained the current working article corpus for analysis.
6. Combined available article title, body, description, summary and related text fields for conservative text-based extraction.
7. Extracted state evidence only when supported by explicit state or associated place-name terms.
8. Extracted time-of-day evidence from explicit wording; records without such wording were retained as Unknown.
9. Created binary indicators for reported rain/wet conditions, fog, night evidence, highway involvement, vehicle types and collision characteristics.
10. Assigned candidate severity conservatively and retained Unknown cases outside the pilot modeling set.
11. Excluded deaths and injuries from predictors to avoid target leakage.
12. Used only records with non-unknown severity and clear single-state evidence in the saved pilot model dataset.

3.2 Severity Distribution

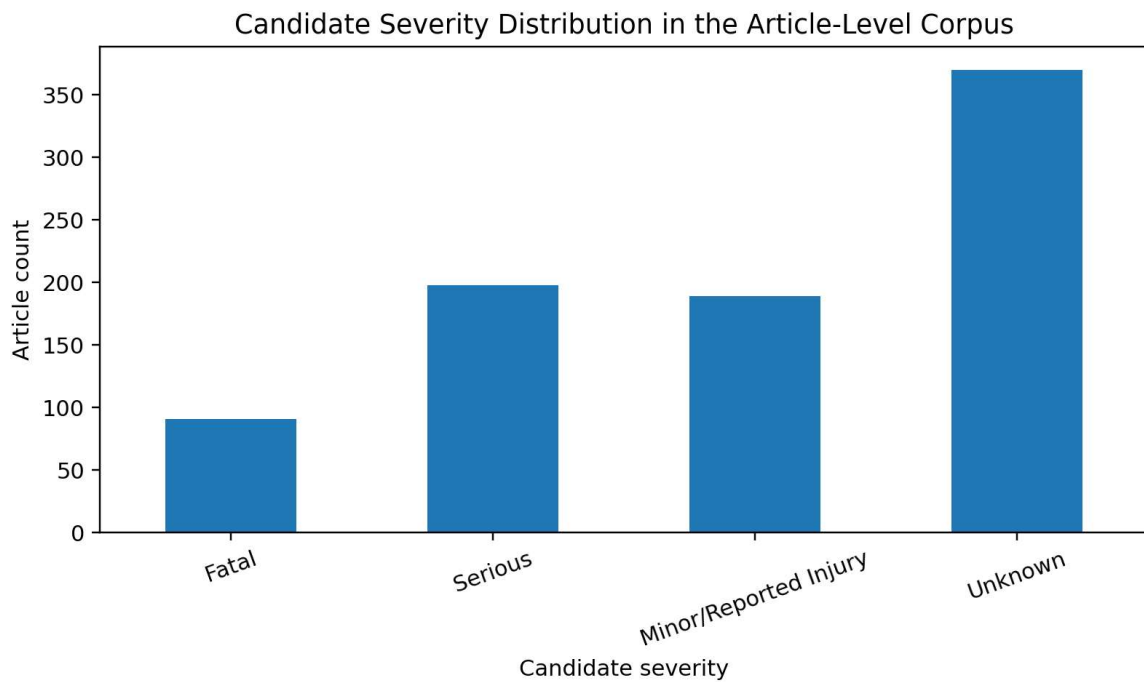


Figure 5. Candidate severity distribution in the article-level corpus

The previously validated article-level extraction identified 91 Fatal, 198 Serious, 189 Minor/Reported Injury and 370 Unknown candidate records. The large Unknown group is important: it demonstrates that news articles do not consistently provide enough evidence for a reliable severity label. For this reason, the final pilot model was trained only on the validated subset rather than treating every scraped article as a labeled accident.

3.3 Location Distribution

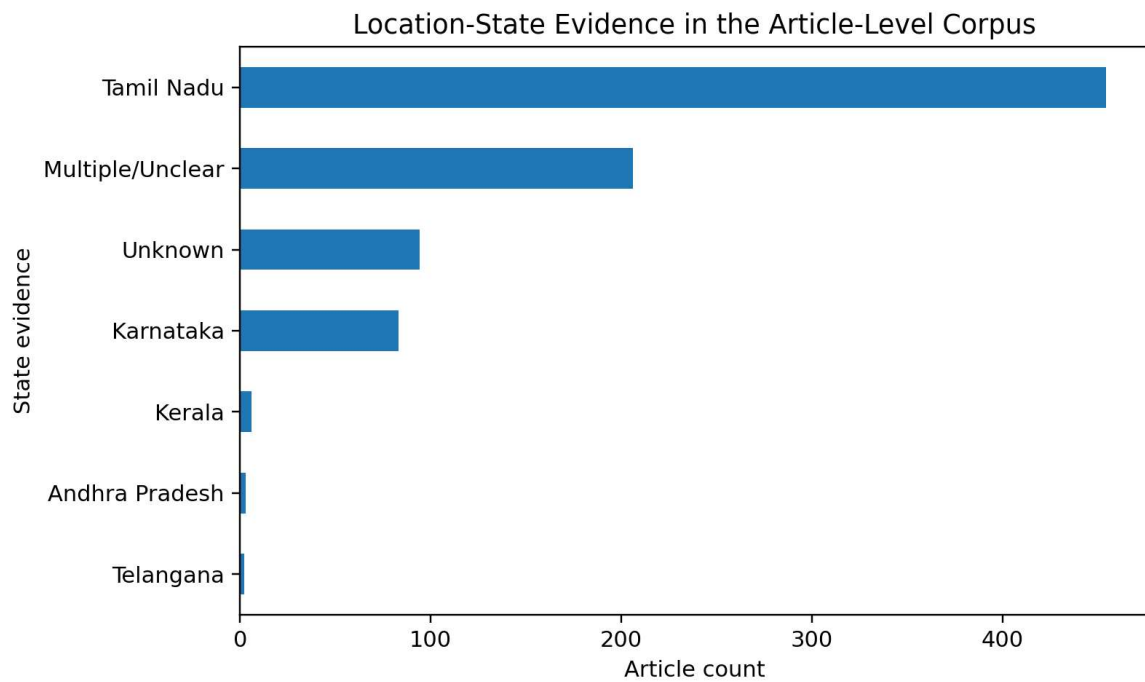


Figure 6. State/location evidence in the article-level corpus

Tamil Nadu provides the largest single-state evidence group, while a substantial number of records remain Multiple/Unclear or Unknown. This distribution reflects the search strategy and reporting coverage; it should not be interpreted as a population estimate of accident frequency across states.

3.4 Time-of-Day Evidence

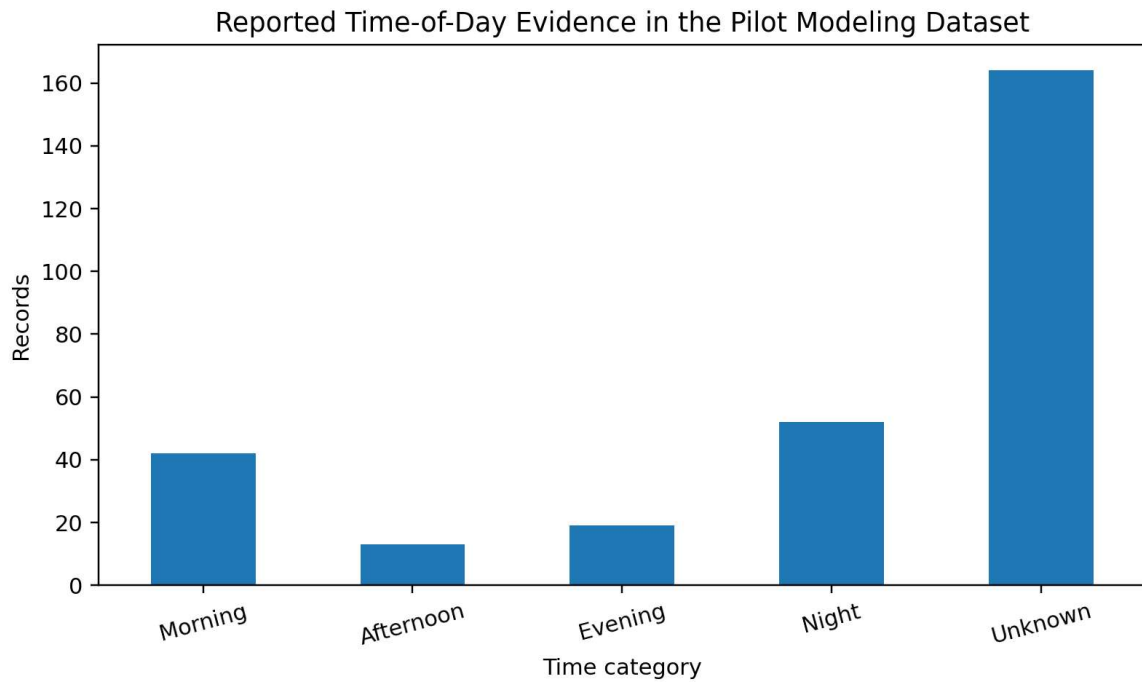


Figure 7. Reported time-of-day evidence in the pilot modeling dataset

Time-of-day evidence is incomplete because many articles do not state the exact time of the accident. Among the 290 pilot records, 52 contain night evidence, 42 morning, 19 evening and 13 afternoon evidence, while 164 remain Unknown. This supports using time as a contextual predictor while recognizing missingness as a limitation.

3.5 Accident-Related Factor Evidence

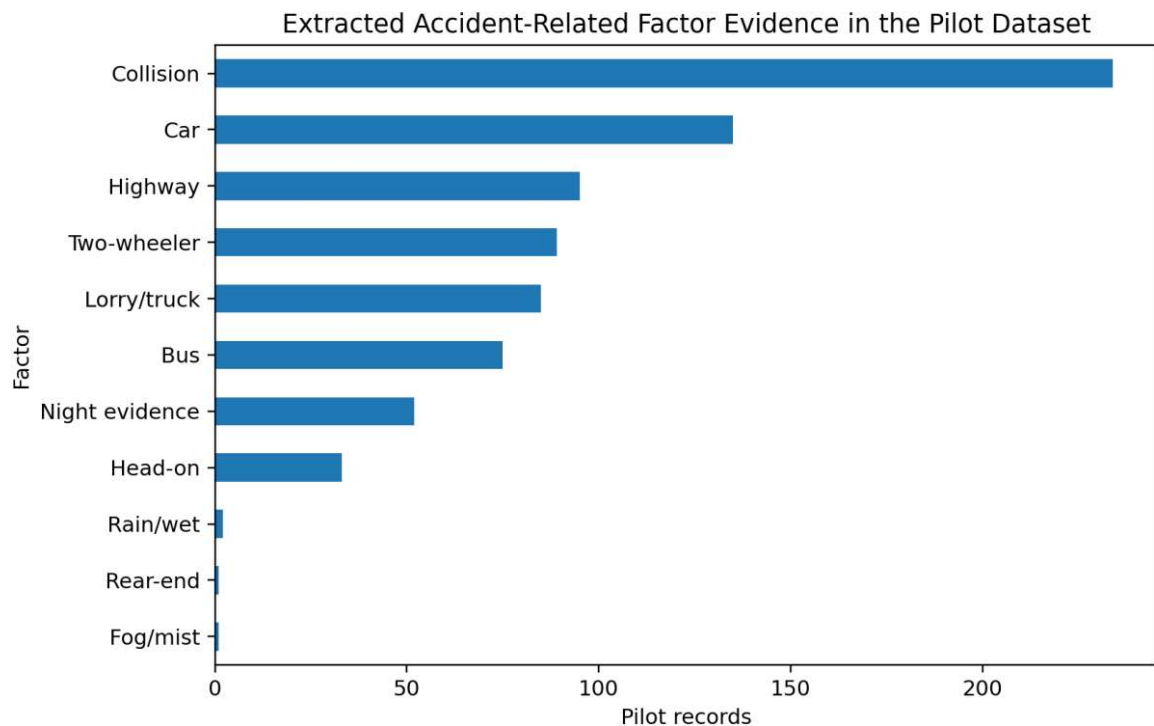


Figure 8. Extracted accident-related factor evidence in the pilot modeling dataset

Collision wording is present in most pilot records, while vehicle indicators such as car, lorry/truck, bus and two-wheeler involvement are also frequently represented. Highway evidence occurs in 95 pilot records. Weather-specific evidence is much sparser: only 2 records contain rain/wet evidence and 1 contains fog/mist evidence in this conservative extraction. This limitation should be considered before drawing strong conclusions about weather effects.

3.6 Pilot Class Distribution

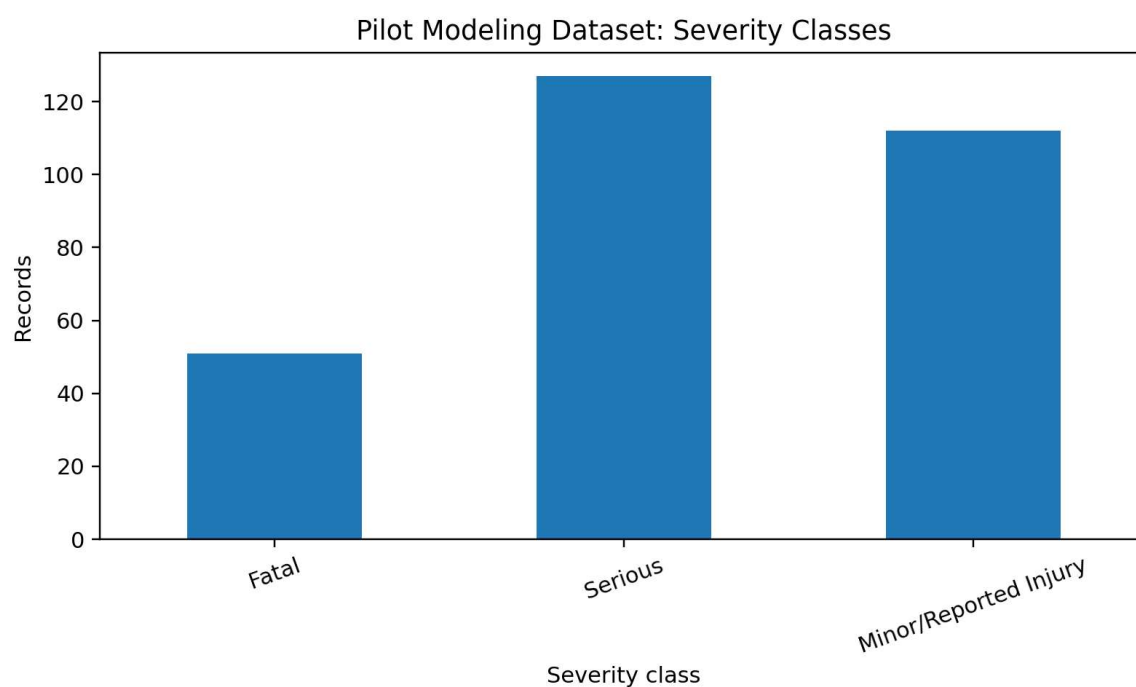


Figure 9. Severity classes in the saved pilot modeling dataset

The pilot modeling dataset contains 290 records: 51 Fatal, 127 Serious and 112 Minor/Reported Injury. This class distribution is the basis for the saved model evaluation and is separate from the larger article-level corpus.

4. Analytics Method and Implementation

The analytical task is multiclass classification: predict a reported severity class from contextual indicators. The method is suitable for Business Analytics because the output can be used as decision-support information for prioritizing road-safety analysis.

Modeling Procedure

13. Use the saved 290-record pilot dataset.
14. Split the data into training and test sets using an 80:20 stratified split with random_state=42.
15. One-hot encode the categorical variables state and time_of_day.
16. Keep binary contextual indicators for weather, road, vehicle and collision evidence.
17. Train four classifiers: Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.
18. Evaluate using accuracy, macro precision, macro recall and macro F1 so that all severity classes contribute to the evaluation.

4.1 Model Results

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	0.362	0.362	0.350	0.345
Decision Tree	0.414	0.425	0.415	0.391
Random Forest	0.397	0.375	0.378	0.375
Gradient Boosting	0.397	0.282	0.321	0.298

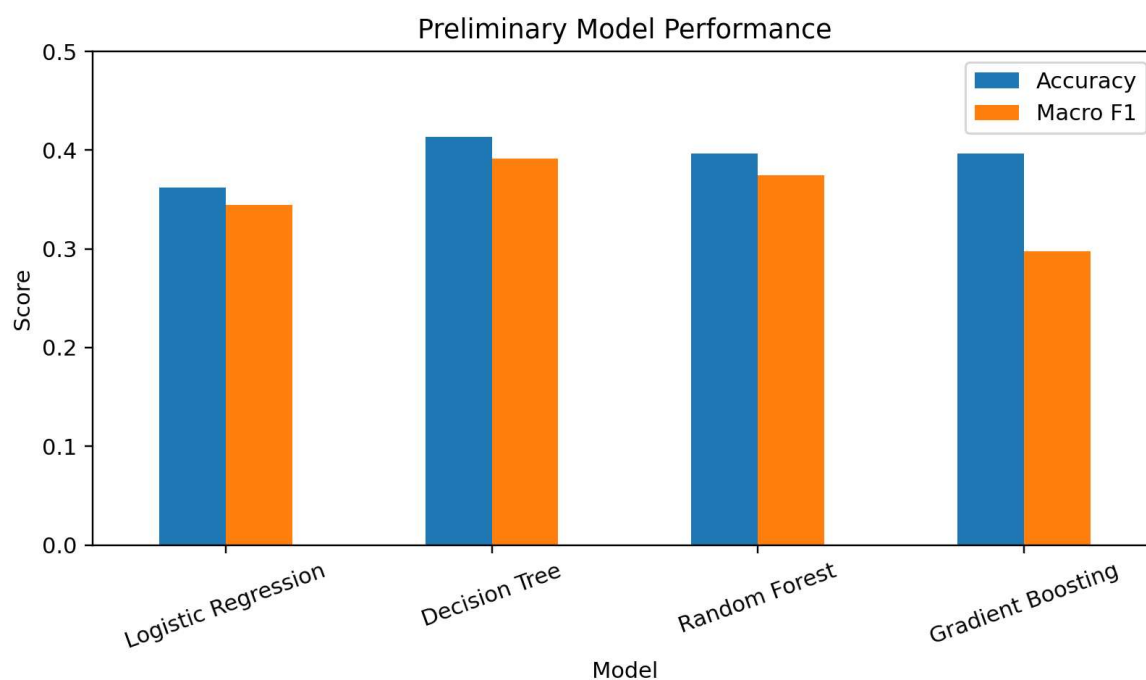


Figure 10. Preliminary model performance on the saved pilot experiment

The Decision Tree produced the highest macro F1 (0.391) and accuracy (0.414) in this saved pilot experiment. This statement is limited to the present dataset and evaluation split; it is not a claim that the Decision Tree is universally superior for road-accident severity prediction.

4.2 Confusion Matrix

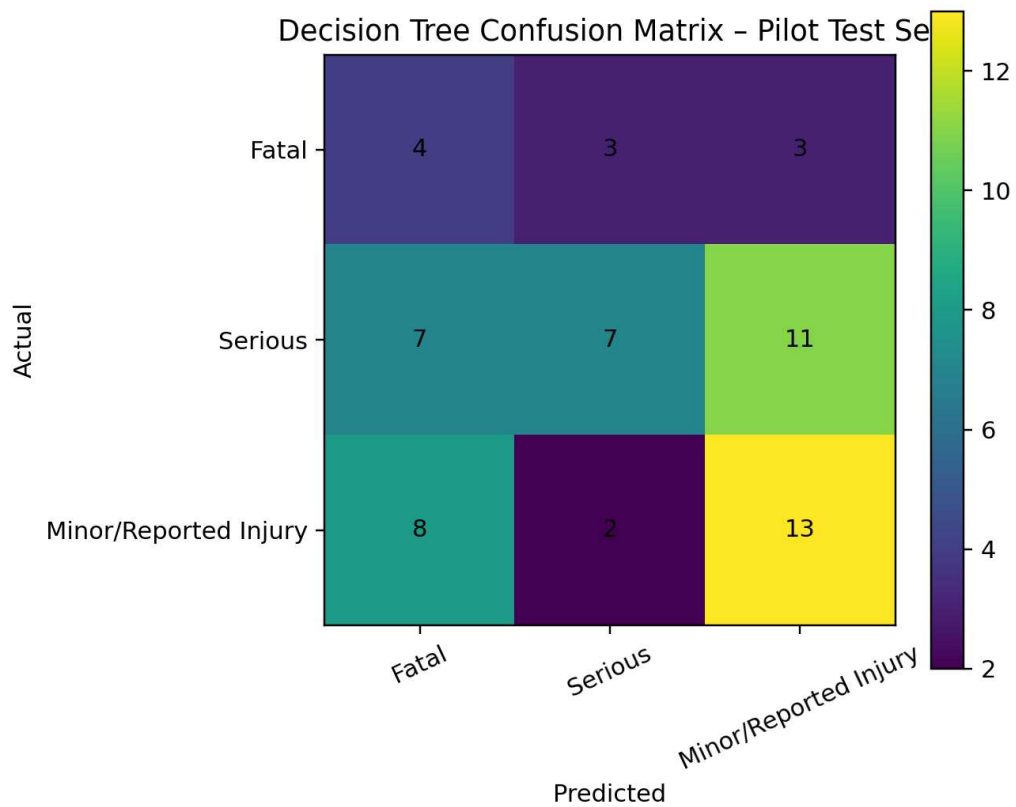


Figure 11. Decision Tree confusion matrix on the pilot test set

The confusion matrix shows that the pilot model makes errors across all three classes. This reinforces the need for better validated accident-event features and a larger, cleaner event-level dataset before operational use.

5. Comparison with State-of-the-Art Methods

Recent road-accident severity studies use statistical and machine-learning approaches with increasing emphasis on weather and traffic context, class imbalance, explainability and practical policy interpretation. The studies below are compared by method, dataset context, evaluation focus and relationship to this case study. Reported scores are not ranked across studies because their datasets and experimental settings differ.

Published study / year	Dataset / context	Method used	Evaluation / focus	Comparison with this work
Zhang et al. (2023)	Mountain-freeway accident severity with dynamic traffic/weather information	Machine-learning severity modeling	Predictive performance with dynamic contextual factors	Supports the use of weather and contextual variables; our source is news-based and broader in road context.
Expressway crash-severity study (2023)	Expressway traffic crash severity	Logistic regression plus explainable/supervised ML classifiers	Severity classification and explainability	Supports comparing conventional and ML classifiers; our pilot also compares multiple classifiers.
Khanum et al. (2025)	Indian highway accident severity	Machine-learning framework	Severity prediction for Indian highways	Provides India-specific methodological context; our data are collected independently through public web scraping.
Phojaem et al. (2026)	Highway-speed motorway injury severity	Comparative injury-severity modeling	Performance, interpretability and policy implications	Highlights the importance of interpretability and policy use; our report similarly treats the model as decision support.

The main methodological difference is the data source. Published studies commonly use structured crash databases, whereas this case study constructs an intermediate accident-report corpus from public webpages. This makes event-level deduplication and validation a central part of the present study.

6. Results, Business Insights and Recommendations

6.1 Main Results

- The current working corpus contains 848 article-level records after the later collection expansion.
- The article-level corpus contains substantial Unknown severity and location ambiguity, demonstrating the importance of validation.
- The saved pilot modeling dataset contains 290 records with three severity classes.
- The Decision Tree achieved the highest pilot macro F1 of 0.391 and accuracy of 0.414 among the four tested models.
- Weather indicators are relatively sparse in the current text extraction, so strong conclusions about weather effects are not justified from this pilot alone.

6.2 Business / Operational Insights

- Road-safety teams can use validated severity patterns to prioritize locations and circumstances for further investigation.
- Time-of-day and highway indicators can support targeted review of patrol coverage, warning signage and safety campaigns.
- Vehicle and collision indicators can help identify recurring accident configurations for engineering or enforcement review.
- A model can serve as a decision-support layer for prioritization, but human review should remain part of operational decisions.
- The collection pipeline can be repeated periodically to build a more current accident-report monitoring dataset.

6.3 Recommendations

19. Perform event-level deduplication so that multiple news reports about the same accident become one validated accident record.
20. Validate accident date and location separately from article publication date.
21. Add structured historical weather information only after the accident location and date are validated.
22. Expand the collection to obtain a larger number of unique accident events across the selected states.
23. Compare additional models and use cross-validation after the event-level dataset is finalized.
24. Retain an audit trail showing why each accident event was included, excluded or marked uncertain.

6.4 Limitations

- News reports are not a random sample of all accidents and may have selection and publication bias.
- Several articles can describe the same accident event.
- Publication date is not necessarily the accident date.
- Weather and road-condition information is often missing or incomplete.
- Text-based extraction can contain ambiguity and location contamination.
- The pilot model uses 290 article-derived records rather than a final one-row-per-accident dataset.
- Therefore the reported 0.391 macro F1 should be interpreted as a preliminary pilot metric, not as final operational model accuracy.

7. Conclusion

This case study demonstrates a complete Business Analytics workflow for studying road-accident severity using publicly accessible web information. The work covers automated data collection through Apify, preprocessing, exploratory analysis, feature construction, supervised classification, model evaluation and practical interpretation.

The current working article corpus contains 848 records, while the saved pilot modeling dataset contains 290 records after conservative severity and location filtering. Among Logistic Regression, Decision Tree, Random Forest and Gradient Boosting, the Decision Tree produced the highest macro F1 in the saved pilot experiment at 0.391, with an accuracy of 0.414.

The most important conclusion is not the model score alone. The analysis shows that data quality and event-level validation are critical when news reports are used as the source. Duplicate reports, incomplete descriptions, ambiguous locations and missing weather information can affect both the target label and predictor variables.

Accordingly, the next stage should convert the article corpus into a validated one-row-per-accident event dataset. Once that process is complete, the same analytics pipeline can be rerun to obtain final accident-level performance and stronger road-safety insights. The present report therefore provides a reproducible preliminary framework while clearly documenting the limitations that must be addressed before operational deployment.

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